```
# 1. Remove the broken versions
!pip uninstall torch torchvision torchaudio -y

# 2. Clear the pip cache to ensure it doesn't just re-install the
broken file
!pip cache purge

# 3. Install the specific stable version for Windows/CUDA
!pip install torch torchvision torchaudio --index-url
https://download.pytorch.org/whl/cu121
```

```
Found existing installation: torch 2.1.2
Uninstalling torch-2.1.2:
  Successfully uninstalled torch-2.1.2

WARNING: Skipping torchvision as it is not installed.
WARNING: Skipping torchaudio as it is not installed.

Files removed: 325
Looking in indexes: https://download.pytorch.org/whl/cu121
Collecting torch
  Downloading https://download.pytorch.org/whl/cu121/torch-
2.4.1%2Bcu121-cp38-cp38-win_amd64.whl (2444.9 MB)
Collecting torchvision
  Downloading https://download-r2.pytorch.org/whl/cu121/torchvision-
0.19.1%2Bcu121-cp38-cp38-win_amd64.whl (5.8 MB)
Collecting torchaudio
  Downloading https://download-r2.pytorch.org/whl/cu121/torchaudio-
2.4.1%2Bcu121-cp38-cp38-win_amd64.whl (4.1 MB)
Requirement already satisfied: fsspec in c:\users\djsce.student\
anaconda3\lib\site-packages (from torch) (0.9.0)
Requirement already satisfied: jinja2 in c:\users\djsce.student\
anaconda3\lib\site-packages (from torch) (2.11.3)
Requirement already satisfied: sympy in c:\users\djsce.student\
anaconda3\lib\site-packages (from torch) (1.8)
Requirement already satisfied: typing-extensions>=4.8.0 in c:\users\
djsce.student\anaconda3\lib\site-packages (from torch) (4.13.2)
Requirement already satisfied: filelock in c:\users\djsce.student\
anaconda3\lib\site-packages (from torch) (3.0.12)
Requirement already satisfied: networkx in c:\users\djsce.student\
anaconda3\lib\site-packages (from torch) (2.5)
Requirement already satisfied: numpy in c:\users\djsce.student\
anaconda3\lib\site-packages (from torchvision) (1.22.4)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in c:\users\
djsce.student\anaconda3\lib\site-packages (from torchvision) (8.2.0)
Requirement already satisfied: MarkupSafe>=0.23 in c:\users\
djsce.student\anaconda3\lib\site-packages (from jinja2->torch) (1.1.1)
Requirement already satisfied: decorator>=4.3.0 in c:\users\
djsce.student\anaconda3\lib\site-packages (from networkx->torch)
(5.0.6)
```

```
Requirement already satisfied: mpmath>=0.19 in c:\users\djsce.student\
anaconda3\lib\site-packages (from sympy->torch) (1.2.1)
Installing collected packages: torch, torchvision, torchaudio
Successfully installed torch-2.4.1+cu121 torchaudio-2.4.1+cu121
torchvision-0.19.1+cu121
```

```
%pip install --force-reinstall torch torchvision torchaudio --index-
url https://download.pytorch.org/whl/cu121
```

```
Looking in indexes: https://download.pytorch.org/whl/cu121
Collecting torch
  Using cached https://download.pytorch.org/whl/cu121/torch-
2.4.1%2Bcu121-cp38-cp38-win_amd64.whl (2444.9 MB)
Collecting torchvision
  Using cached https://download-r2.pytorch.org/whl/cu121/torchvision-
0.19.1%2Bcu121-cp38-cp38-win_amd64.whl (5.8 MB)
Collecting torchaudio
  Using cached https://download-r2.pytorch.org/whl/cu121/torchaudio-
2.4.1%2Bcu121-cp38-cp38-win_amd64.whl (4.1 MB)
Collecting fsspec
  Downloading fsspec-2025.3.0-py3-none-any.whl (193 kB)
Note: you may need to restart the kernel to use updated
packages.Collecting filelock
```

```
ERROR: Could not install packages due to an OSError: [WinError 5]
Access is denied: 'C:\\Users\\djsce.student\\Anaconda3\\Lib\\site-
packages\\~orch\\lib\\asmjit.dll'
Consider using the `--user` option or check the permissions.
```

```
  Downloading filelock-3.16.1-py3-none-any.whl (16 kB)
Collecting jinja2
  Downloading https://download.pytorch.org/whl/jinja2-3.1.6-py3-none-
any.whl (134 kB)
Collecting sympy
  Downloading sympy-1.13.3-py3-none-any.whl (6.2 MB)
Collecting typing-extensions>=4.8.0
  Downloading https://download.pytorch.org/whl/typing_extensions-
4.15.0-py3-none-any.whl (44 kB)
Collecting networkx
  Downloading networkx-3.1-py3-none-any.whl (2.1 MB)
Collecting typing-extensions>=4.8.0
  Downloading https://download.pytorch.org/whl/typing_extensions-
4.14.0-py3-none-any.whl (43 kB)
  Downloading https://download.pytorch.org/whl/typing_extensions-
4.12.2-py3-none-any.whl (37 kB)
Collecting numpy
  Downloading numpy-1.24.4-cp38-cp38-win_amd64.whl (14.9 MB)
Collecting pillow!=8.3.*,>=5.3.0
```

```
  Downloading pillow-10.4.0-cp38-cp38-win_amd64.whl (2.6 MB)
Collecting MarkupSafe>=2.0
  Downloading https://download.pytorch.org/whl/MarkupSafe-2.1.5-cp38-
cp38-win_amd64.whl (17 kB)
Collecting mpmath<1.4,>=1.1.0
  Downloading mpmath-1.3.0-py3-none-any.whl (536 kB)
Installing collected packages: mpmath, MarkupSafe, typing-extensions,
sympy, networkx, jinja2, fsspec, filelock, torch, pillow, numpy,
torchvision, torchaudio
  Attempting uninstall: mpmath
    Found existing installation: mpmath 1.2.1
    Uninstalling mpmath-1.2.1:
      Successfully uninstalled mpmath-1.2.1
  Attempting uninstall: MarkupSafe
    Found existing installation: MarkupSafe 1.1.1
    Uninstalling MarkupSafe-1.1.1:
      Successfully uninstalled MarkupSafe-1.1.1
  Attempting uninstall: typing-extensions
    Found existing installation: typing-extensions 4.13.2
    Uninstalling typing-extensions-4.13.2:
      Successfully uninstalled typing-extensions-4.13.2
  Attempting uninstall: sympy
    Found existing installation: sympy 1.8
    Uninstalling sympy-1.8:
      Successfully uninstalled sympy-1.8
  Attempting uninstall: networkx
    Found existing installation: networkx 2.5
    Uninstalling networkx-2.5:
      Successfully uninstalled networkx-2.5
  Attempting uninstall: jinja2
    Found existing installation: Jinja2 2.11.3
    Uninstalling Jinja2-2.11.3:
      Successfully uninstalled Jinja2-2.11.3
  Attempting uninstall: fsspec
    Found existing installation: fsspec 0.9.0
    Uninstalling fsspec-0.9.0:
      Successfully uninstalled fsspec-0.9.0
  Attempting uninstall: filelock
    Found existing installation: filelock 3.0.12
    Uninstalling filelock-3.0.12:
      Successfully uninstalled filelock-3.0.12
  Attempting uninstall: torch
    Found existing installation: torch 2.4.1+cu121
    Uninstalling torch-2.4.1+cu121:
      Successfully uninstalled torch-2.4.1+cu121
```

```python
import torch

if torch.cuda.is_available():
    print(f"□ Success! GPU detected: {torch.cuda.get_device_name(0)}")
```

```
    device = torch.device("cuda")
else:
    print("⚠ Still on CPU. Checking for drivers...")
    # This checks if Windows actually sees an NVIDIA card
    import subprocess
    try:
        print(subprocess.check_output(['nvidia-smi']).decode('utf-8'))
    except:
        print("NVIDIA Drivers not found on this system.")
    device = torch.device("cpu")
```

```
⚠ Still on CPU. Checking for drivers...
NVIDIA Drivers not found on this system.
```

```
import os, glob

DATA_DIR  = r"D:\huhuhuhuh\yay"
TRAIN_DIR = os.path.join(DATA_DIR, "train")
TEST_DIR  = os.path.join(DATA_DIR, "test")

csvs = glob.glob(os.path.join(DATA_DIR, "*.csv"))
if not csvs:
    raise FileNotFoundError("No CSV found. Put your bounding box csv
inside your DATA_DIR")
CSV_PATH = csvs[0]

print("📂 TRAIN_DIR:", TRAIN_DIR)
print("📂 TEST_DIR :", TEST_DIR)
print("📄 CSV_PATH :", CSV_PATH)
```

```
📂 TRAIN_DIR: D:\huhuhuhuh\yay\train
📂 TEST_DIR : D:\huhuhuhuh\yay\test
📄 CSV_PATH : D:\huhuhuhuh\yay\_annotations.csv
```

```
train_count = len(glob.glob(os.path.join(TRAIN_DIR, "*.*")))
test_count = len(glob.glob(os.path.join(TEST_DIR, "*.*")))

print(f"📸 Train images: {train_count}")
print(f"📸 Test images: {test_count}")
```

```
📸 Train images: 102
📸 Test images: 13
```

```
import pandas as pd

df = pd.read_csv(r"D:\huhuhuhuh\yay\_annotations.csv")

print("CSV columns:", df.columns.tolist())
print(df.head())
```

```
CSV columns: ['filename', 'width', 'height', 'class', 'xmin', 'ymin',
'xmax', 'ymax']
```

```
                                                filename  width  height
class  \
0  frame_9538_jpg.rf.03110592999f1dbf6819eea2a816...    640     640
Car
1  frame_3390_jpg.rf.0c8f2506e30861bdd7039fadd515...    640     640
Car
2  frame_3390_jpg.rf.0c8f2506e30861bdd7039fadd515...    640     640
Truck
3  frame_5968_jpg.rf.0df1e2610361d00c9804ea25b39f...    640     640
Car
4  frame_5968_jpg.rf.0df1e2610361d00c9804ea25b39f...    640     640
Car

   xmin  ymin  xmax  ymax
0   368   110   475   161
1   392   145   462   183
2   446   132   478   153
3   175   362   276   433
4   388   269   438   307
```

```
possible_file_cols = ["image", "image_id", "filename", "file", "img",
"img_name"]
possible_xmin = ["xmin", "x_min", "x1", "left"]
possible_ymin = ["ymin", "y_min", "y1", "top"]
possible_xmax = ["xmax", "x_max", "x2", "right"]
possible_ymax = ["ymax", "y_max", "y2", "bottom"]

def pick_col(possible):
    for c in possible:
        if c in df.columns:
            return c
    return None

file_col = pick_col(possible_file_cols)
xmin_col = pick_col(possible_xmin)
ymin_col = pick_col(possible_ymin)
xmax_col = pick_col(possible_xmax)
ymax_col = pick_col(possible_ymax)

if None in [file_col, xmin_col, ymin_col, xmax_col, ymax_col]:
    raise ValueError("CSV columns not detected properly")

print("Using:", file_col, xmin_col, ymin_col, xmax_col, ymax_col)
```

```
Using: filename xmin ymin xmax ymax
```

```
import numpy as np
from PIL import Image
import torch
from torch.utils.data import Dataset, DataLoader
import torchvision
```

```python
from torchvision.transforms import functional as F
from torchvision.models.detection.faster_rcnn import FastRCNNPredictor


class VehicleDataset(Dataset):

    def __init__(self, images_dir, df):
        self.images_dir = images_dir
        self.df = df.copy()
        self.image_ids = sorted(self.df[file_col].unique().tolist())

    def __len__(self):
        return len(self.image_ids)

    def _resolve_path(self, name):
        p = os.path.join(self.images_dir, str(name))
        if os.path.exists(p):
            return p

        if not os.path.splitext(str(name))[1]:
            for ext in [".jpg", ".jpeg", ".png"]:
                p2 = os.path.join(self.images_dir, str(name) + ext)
                if os.path.exists(p2):
                    return p2

        m = glob.glob(os.path.join(self.images_dir, str(name) + ".*"))

        if m:
            return m[0]

        raise FileNotFoundError(f"Image not found: {name}")

    def __getitem__(self, idx):

        img_id = self.image_ids[idx]
        rows = self.df[self.df[file_col] == img_id]

        img_path = self._resolve_path(img_id)
        img = Image.open(img_path).convert("RGB")

        boxes = rows[[xmin_col, ymin_col, xmax_col,
ymax_col]].values.astype(np.float32)

        valid = (boxes[:, 2] > boxes[:, 0]) & (boxes[:, 3] > boxes[:,
1])
        boxes = boxes[valid]

        labels = np.ones((len(boxes),), dtype=np.int64)

        target = {
            "boxes": torch.tensor(boxes, dtype=torch.float32),
```

```
            "labels": torch.tensor(labels, dtype=torch.int64),
            "image_id": torch.tensor([idx]),
        }

        img = F.to_tensor(img)
        return img, target

def collate_fn(batch):
    return tuple(zip(*batch))

def get_model(num_classes=2):
    model =
torchvision.models.detection.fasterrcnn_resnet50_fpn(weights="DEFAULT"
)
    in_features = model.roi_heads.box_predictor.cls_score.in_features
    model.roi_heads.box_predictor = FastRCNNPredictor(in_features,
num_classes)
    return model

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Device:", device)

dataset = VehicleDataset(TRAIN_DIR, df)

n = len(dataset)
train_len = int(0.9 * n)
val_len = n - train_len
train_ds, val_ds = torch.utils.data.random_split(dataset, [train_len,
val_len])

train_loader = DataLoader(train_ds, batch_size=2, shuffle=True,
num_workers=0, collate_fn=collate_fn)
val_loader   = DataLoader(val_ds, batch_size=2, shuffle=False,
num_workers=0, collate_fn=collate_fn)

model = get_model(num_classes=2).to(device)
```

```
Device: cpu

Downloading:
"https://download.pytorch.org/models/fasterrcnn_resnet50_fpn_coco-
258fb6c6.pth" to C:\Users\djsce.student/.cache\torch\hub\checkpoints\
fasterrcnn_resnet50_fpn_coco-258fb6c6.pth
100%|██████████| 160M/160M [00:01<00:00, 89.9MB/s]
```

```
params = [p for p in model.parameters() if p.requires_grad]
optimizer = torch.optim.SGD(params, lr=0.005, momentum=0.9,
weight_decay=0.0005)

EPOCHS = 5
```

```
for epoch in range(EPOCHS):

    model.train()
    total_loss = 0

    for images, targets in train_loader:

        images = [img.to(device) for img in images]
        targets = [{k: v.to(device) for k, v in t.items()} for t in
targets]

        loss_dict = model(images, targets)
        losses = sum(loss for loss in loss_dict.values())

        optimizer.zero_grad()
        losses.backward()
        optimizer.step()

        total_loss += losses.item()

    print(f"Epoch {epoch+1}/{EPOCHS}, Loss:
{total_loss/len(train_loader):.4f}")
```

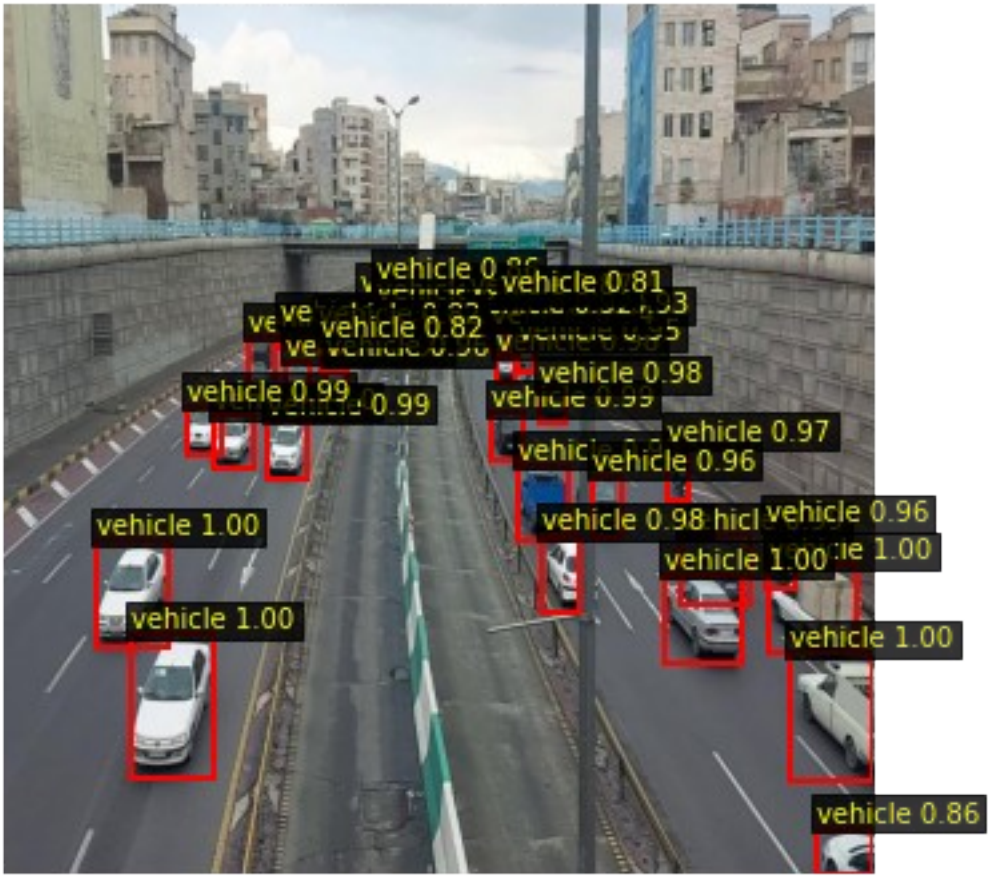
```
Epoch 1/5, Loss: 0.7788
Epoch 2/5, Loss: 0.5252
Epoch 3/5, Loss: 0.4519
Epoch 4/5, Loss: 0.4016
Epoch 5/5, Loss: 0.3748
```

```
import matplotlib.pyplot as plt
import matplotlib.patches as patches
from torchvision.ops import nms

def get_test_files(test_dir):

    files = []
    for ext in ["*.jpg", "*.jpeg", "*.png"]:
        files += glob.glob(os.path.join(test_dir, ext))

    return sorted(files)

def show_predictions(image_path, score_thresh=0.8, nms_thresh=0.3):

    model.eval()
    img_pil = Image.open(image_path).convert("RGB")
    img_t = F.to_tensor(img_pil).to(device)

    with torch.no_grad():

        pred = model([img_t])[0]
```

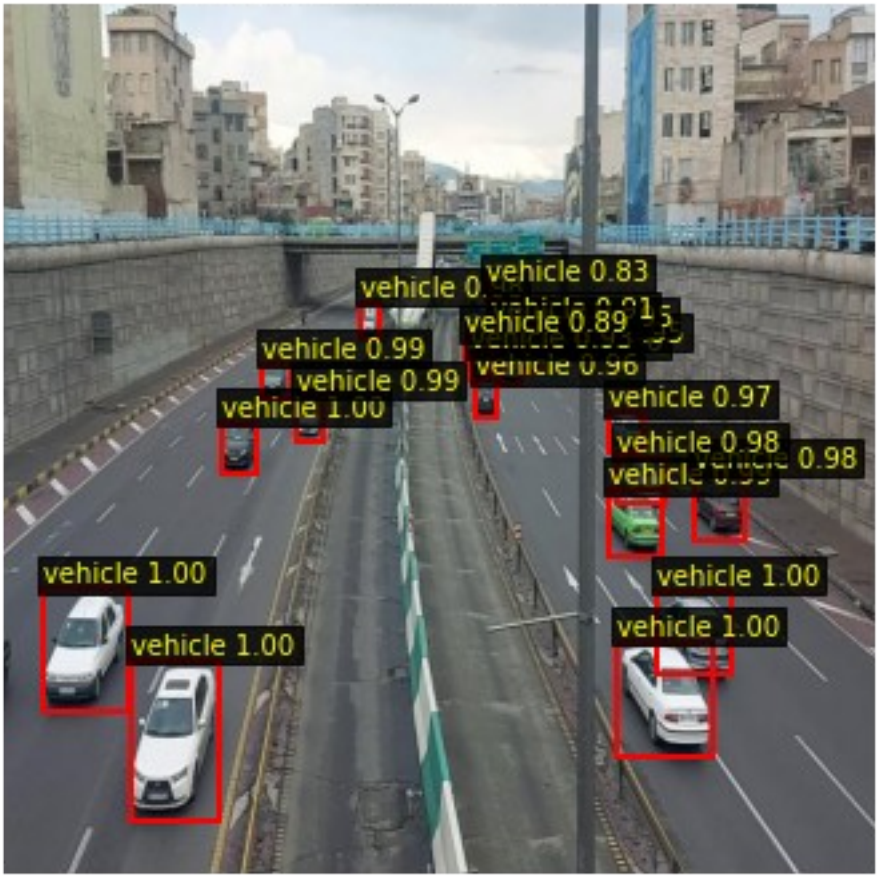
```
    boxes = pred["boxes"]
    scores = pred["scores"]

    keep = scores >= score_thresh
    boxes = boxes[keep]
    scores = scores[keep]

    if len(boxes) > 0:

        keep_idx = nms(boxes, scores, nms_thresh)
        boxes = boxes[keep_idx]
        scores = scores[keep_idx]

    boxes = boxes.cpu().numpy()
    scores = scores.cpu().numpy()

    fig, ax = plt.subplots(figsize=(10, 6))
    ax.imshow(img_pil)
    ax.axis("off")
    ax.set_title(os.path.basename(image_path))

    for (x1, y1, x2, y2), s in zip(boxes, scores):

        rect = patches.Rectangle((x1, y1), x2-x1, y2-y1,
                                 linewidth=2, edgecolor="red",
facecolor="none")

        ax.add_patch(rect)
        ax.text(x1, max(0, y1-5), f"vehicle {s:.2f}",

                color="yellow",
                bbox=dict(facecolor="black", alpha=0.8, pad=2))

    plt.show()

test_files = get_test_files(TEST_DIR)
print("Test images found:", len(test_files))

for p in test_files[-5:]:

    show_predictions(p, score_thresh=0.8, nms_thresh=0.3)
```

```
Test images found: 13
```

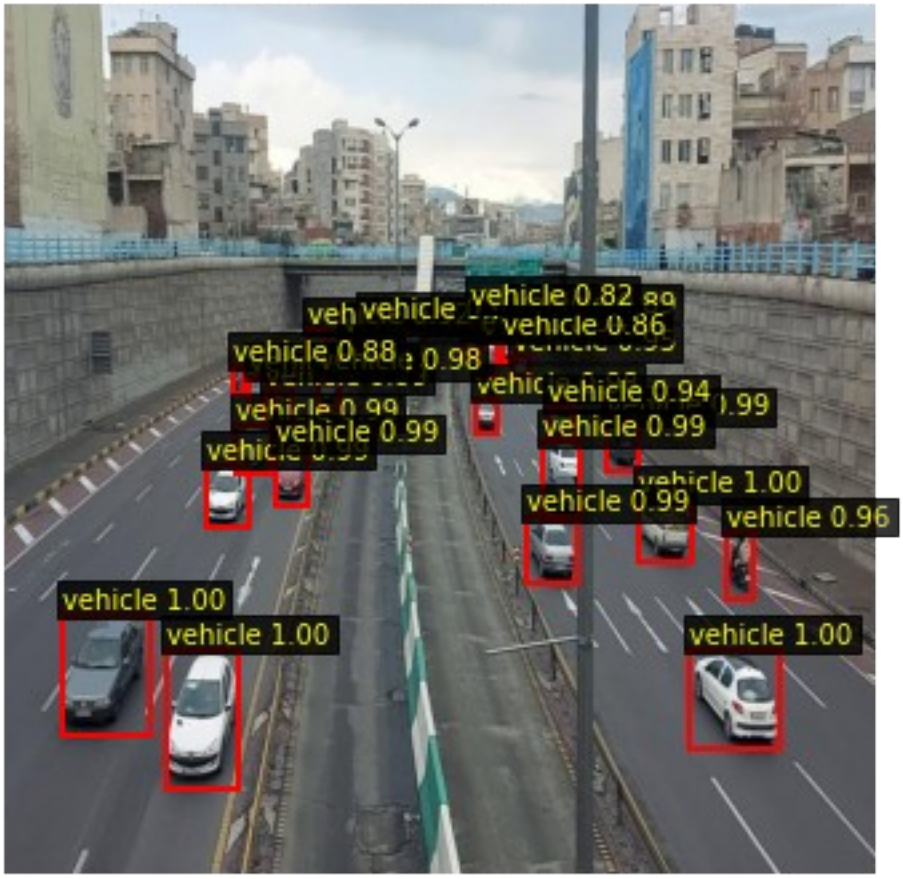
frame_7929_jpg.rf.02b89050254d7e4c8c8995125a555cfe.jpg



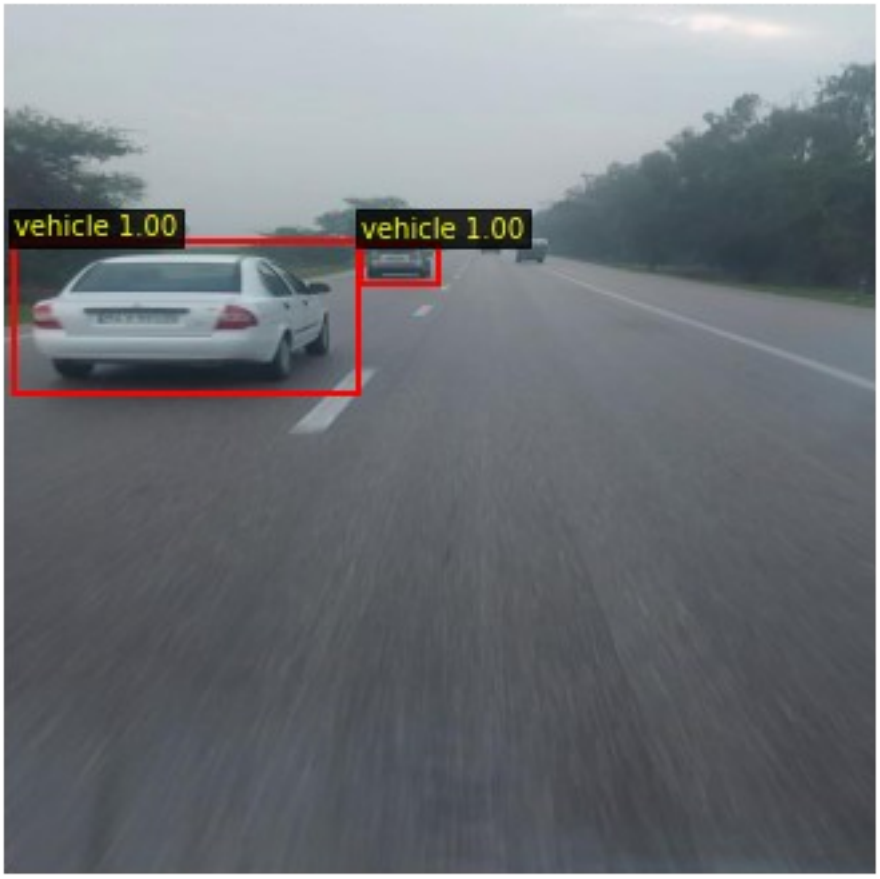
frame_8097_jpg.rf.762bcc82c7b05b2ebcabc219815dca50.jpg



frame_9055_jpg.rf.067db6b83a7de9859840f6ca2484d373.jpg



frame_9106_jpg.rf.edaac933241f83e08bf00984c0e0c85f.jpg



frame_9486_jpg.rf.2280c498ddc7234aff225ca73be23450.jpg